

Distant animal

Moving in the world frame: **the horse**. Staying in place: 9 trees.

The horse moves about 1.4 m in the world frame at an average speed of 0.23 m/s; it accelerates over the clip.

Question

Right answer: C

In the world frame (ignoring camera motion), what was the average speed (in meters per second) of **the horse** between frame 0 and frame 32 of 34?

A. 0.29 B. 0.39 **C. 0.21** D. 0.16

ground truth: 0.22 m/s

Qwen3-VL-8B · frames only

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<evidence>
object: the horse
position_frame_0_m: (0.1, -0.0, 13.0)
position_frame_32_m: (2.6, 0.0, 10.4)
displacement_m: 3.6
time_s: 6.41
speed_mps: 0.56
</evidence>
<answer>B</answer>
```

Ours · frames + trajectory tokens

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<evidence>
object: the horse
position_frame_0_m: (0.1, 0.2, 5.1)
position_frame_32_m: (1.4, 0.1, 4.7)
displacement_m: 1.4
time_s: 6.41
speed_mps: 0.22
</evidence>
<answer>C</answer>
```